Type MS · Link a nutrient to its function, source and deficiency disease

Answer 1 of 13 · Question 2 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

2(a)	Any two from: carbohydrates/proteins/lipids/mineral salts (or named example)/fibre;;	Ignore named examples (1 mark for each two correct)	**(2)**

Question Number	Answer	Additional guidance	Mark
2(b)	Any two from: • lack of it causes scurvy; • (lack of vitamin C causes) damage to membranes/bleeding gums; • makes connective tissue/maintains/strengthens cell linings/gums/sticks/joins cells/tissues together;		**(2)**

Question Number	Answer	Additional guidance	Mark
2(c)(i)	Any five from: • Add fresh juice/long-life juice (to a test tube); • add DCPIP drop by drop/slowly; • until blue colour disappears/solution turns colourless; • count number of drops/measure volume of DCPIP added; • repeat (to obtain mean); • compare volumes of DCPIP added;		**(5)**

Question Number	Answer	Additional guidance	Mark
2(c)(ii)	Any two from: • volume of juice; • temperature; • concentration DCPIP;	Allow amount for mp1. Ignore concentration	**(2)**

Total 11 marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 1 of 13 · Question 2 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 2 of 13 · Question 4 part c · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

4 (a)	$\frac{30\,000}{100}$ x 40; 12 000;	Full marks for final correct answer ecf	1 1
(b)	a microorganism/named microorganism that causes disease;		1
(c)	It cannot be transmitted from one person to the next		1
(d)	• use a sample of people/use two groups of people; • feed each sample/group of people a different diet/feed one group a diet with vitamin B and the other without vitamin B; • monitor health/compare (health) of two groups;		1 1 1
(e)	• breakdown of cartilage/cartilage not formed properly/damaged; • bones rub together/more friction between bones/less shock absorption;		1 1

Total question 4 = 9 marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 2 of 13 · Question 4 part c · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 3 of 13 · Question 1 part b.i · 1 mark

2020 JAN · Paper 01 · find part b.i in this mark scheme extract

number				
1	(a)	**The only correct answer is C** to slow down the growth of bacteria *A is incorrect because taste is personal preference* *B is incorrect because it does not kill the bacteria* *D is incorrect because the appearance is not affected*		1
	(b) (i)	protein		1
	(ii)	**The only correct answer is D** it does not contain energy from fat *A is incorrect because vitamins have no effect on body mass* *B is incorrect because cholesterol has no effect on body mass* *C is incorrect because it does contain energy from carbohydrates*		1
	(c)	• add Benedict's solution (to a sample of yogurt) (1) • heat gently (1) • brick red colour indicates the presence of sugar (1)	R if adding to glucose/reducing sugar Allow yellow, green, orange	3

Total for Question 1 = 6 marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 4 of 13 · Question 4 part c.i · 1 mark

2020 JAN · Paper 02 · find part c.i in this mark scheme extract

Number				
4	(a)(i)	pinworms/eggs in faeces/nutritional deficiency/named deficiency/weight loss/malnutrition(1)	Accept fatigue	1
	(ii)	Any two from: • itching transfers eggs (from anus) to fingers/hand(1) • eggs transferred from fingers to other surfaces/people(1) • eggs ingested when other people touch (contaminated) surfaces (and put hands in their mouth)(1)		2
	(b)	Any two from: • correct temperature/pH/provides warmth(1) • plenty of nutrients(1) • no predators(1)	Allow food	2
	(c) (i)	lacking in one or more nutrients/vitamins/minerals / other named nutrient(1)		1
	(ii)	any named nutritional deficiency such as anaemia/scurvy	Allow named nutrient e.g. protein deficiency	1
	(iii)	• pinworms absorb nutrients/named nutrient(1) • less nutrients absorbed/pass into blood/body/cells receive less nutrients(1)	Allow food	2
	(d)	Reference to eggs/pinworms on transparent tape (where they can be seen)		1
	(e)	Any two from: • refrain from itching/scratching (anus)(1) • use medication (to kill pinworms)(1) • don't share towels/clothing(1) • wash bedclothes/linen regularly/do not shake bed linen(1) • don't bite nails/keep fingernails short / clean/wash hands (after using toilet)(1) • don't allow children to bathe together(1)		2

Total for Question 4 = 12 marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 5 of 13 · Question 4 part c.ii · 1 mark

2020 JAN · Paper 02 · find part c.ii in this mark scheme extract

Number				
4	(a)(i)	pinworms/eggs in faeces/nutritional deficiency/named deficiency/weight loss/malnutrition(1)	Accept fatigue	1
	(ii)	Any two from: • itching transfers eggs (from anus) to fingers/hand(1) • eggs transferred from fingers to other surfaces/people(1) • eggs ingested when other people touch (contaminated) surfaces (and put hands in their mouth)(1)		2
	(b)	Any two from: • correct temperature/pH/provides warmth(1) • plenty of nutrients(1) • no predators(1)	Allow food	2
	(c) (i)	lacking in one or more nutrients/vitamins/minerals / other named nutrient(1)		1
	(ii)	any named nutritional deficiency such as anaemia/scurvy	Allow named nutrient e.g. protein deficiency	1
	(iii)	• pinworms absorb nutrients/named nutrient(1) • less nutrients absorbed/pass into blood/body/cells receive less nutrients(1)	Allow food	2
	(d)	Reference to eggs/pinworms on transparent tape (where they can be seen)		1
	(e)	Any two from: • refrain from itching/scratching (anus)(1) • use medication (to kill pinworms)(1) • don't share towels/clothing(1) • wash bedclothes/linen regularly/do not shake bed linen(1) • don't bite nails/keep fingernails short / clean/wash hands (after using toilet)(1) • don't allow children to bathe together(1)		2

Total for Question 4 = 12 marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 6 of 13 · Question 4 part a · 2 marks

2020 NOV · Paper 01 (regional R) · find part a in this mark scheme extract

4 (a)	• ankles(1) • protein(1)		2
(b) (i)	• axes labels(1) • axes scales(1) • correctly plotted bars(1)	Max 2 marks for line graph	3
(ii)	improved/varied/balanced diet/education (in nutrition) / international aid/less protein in diet in 2010(1)		1
(iii)	• 95 ÷ 1978 (x 100) (1) • 4.8(%)	Full marks for correct final answer	2

Total for Question 4 = 8 marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 7 of 13 · Question 4 part b.ii · 1 mark

2020 NOV · Paper 01 (regional R) · find part b.ii in this mark scheme extract

4 (a)	• ankles(1) • protein(1)		2
(b) (i)	• axes labels(1) • axes scales(1) • correctly plotted bars(1)	Max 2 marks for line graph	3
(ii)	improved/varied/balanced diet/education (in nutrition) / international aid/less protein in diet in 2010(1)		1
(iii)	• 95 ÷ 1978 (x 100) (1) • 4.8(%)	Full marks for correct final answer	2

Total for Question 4 = 8 marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 8 of 13 · Question 5 part a.i · 1 mark

2020 NOV · Paper 01 (regional R) · find part a.i in this mark scheme extract

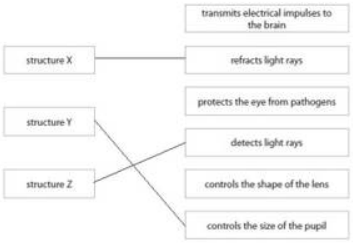
5 (a)	(i)	**C** green vegetables and red meat		1
	(ii)	• (iron) needed for production of haemoglobin(1) • (haemoglobin/red blood cell) binds to/transports oxygen(1) • oxygen needed for (aerobic) respiration/energy(1) • for growth of fetus(1)		4
(b)	(i)	Any four from: • take temperature of water (at the start) (1) • reference to measuring mass of food(1) • place food on holder(1) • burn food(1) • reference to time (for burning food)(1) • take temperature of water after burning/once food is completely burnt/measure change in temperature(1)	Allow amount for mass Allow food is heated/set alight	4
	(ii)	Any three from: • add a lid(1) • reference to stirring/even heat distribution(1) • insulate the copper can/move food closer (to the can/water) (1) • reference to preventing heat loss(1)	Allow cover the beaker	3

Total for question 5 = 12 marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 9 of 13 · Question 1 part b.iv · 1 mark

2023 JAN · Paper 01 · extract 1 of 2 · find part b.iv in this mark scheme extract

1	(a)	(i)			3
		(ii)	A; B, C and D do not describe stereoscopic vision.		1
	(b)	(i)	build-up of protein/cloudy/opaque; in the lens;		2
		(ii)	surgery/operation to replace the lens/lens replacement;		1
		(iii)	39 000000 x 43 ÷ 100; 16.77 million/16.8 million/16 770 000;		2
		(iv)	vitamin A;		1

Total 10 marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 9 of 13 · Question 1 part b.iv · 1 mark

2023 JAN · Paper 01 · extract 2 of 2 · find part b.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 10 of 13 · Question 4 part d · 5 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

Question number				
4	(a)	• correct plots;;; • key/labels;		4
	(b)	• correct components/nutrients/list of components min of five; • in correct amounts/proportions;		2
	(c)	• 75; • 15 • = 5:1;	ECF	2
	(d)	• contain protein • needed for growth/repair/,muscles/tissues; • area A girl has low(er) levels/area B girl has high(er) levels; • less/poor growth/area B girl good growth; • kwashiorkor;		5

Total 13 marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 10 of 13 · Question 4 part d · 5 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 11 of 13 · Question 3 part a.iv · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.iv in this mark scheme extract

3 (a) (i)	contains all nutrients needed/ correct balance of nutrients needed/ correct proportion of nutrients; to maintain health;	If groups of nutrients given all must be named	(2)
(ii)	males need more energy than females; 3000 kJ; greater muscle mass; males are likely to be more active/ higher metabolic rate:	Allow reverse for female	(3)
(iii)	iron needed to form haemoglobin; to replace haemoglobin lost during menstruation/red blood cells are lost during menstruation;		(2)
(iv)	vitamin A;		(1)
(v)	In 250g red meat - 2.5 x 2.5 = 6.25; In 200g broccoli - 2 x 0.7 = 1.4; 12 - 7.65; 4.35 (mg);	Full marks for correct final answer ecf for mp3 and mp4	(4)
(b) (i)	bleeding gums/anaemia/bruising/poor wound healing;	Allow other valid symptoms	(1)
(ii)	more fruit/vegetables eaten in diet/increased intake of vitamin C/ vitamin c supplement;		(1)

Total for Question 3 = 14 marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 12 of 13 · Question 3 part b.i · 1 mark

2024 JUN · Paper 01 (regional R) · find part b.i in this mark scheme extract

3 (a) (i)	contains all nutrients needed/ correct balance of nutrients needed/ correct proportion of nutrients; to maintain health;	If groups of nutrients given all must be named	(2)
(ii)	males need more energy than females; 3000 kJ; greater muscle mass; males are likely to be more active/ higher metabolic rate:	Allow reverse for female	(3)
(iii)	iron needed to form haemoglobin; to replace haemoglobin lost during menstruation/red blood cells are lost during menstruation;		(2)
(iv)	vitamin A;		(1)
(v)	In 250g red meat - 2.5 x 2.5 = 6.25; In 200g broccoli - 2 x 0.7 = 1.4; 12 - 7.65; 4.35 (mg);	Full marks for correct final answer ecf for mp3 and mp4	(4)
(b) (i)	bleeding gums/anaemia/bruising/poor wound healing;	Allow other valid symptoms	(1)
(ii)	more fruit/vegetables eaten in diet/increased intake of vitamin C/ vitamin c supplement;		(1)

Total for Question 3 = 14 marks

Type MS · Link a nutrient to its function, source and deficiency disease

Answer 13 of 13 · Question 3 part b.ii · 1 mark

2024 JUN · Paper 01 (regional R) · find part b.ii in this mark scheme extract

3 (a) (i)	contains all nutrients needed/ correct balance of nutrients needed/ correct proportion of nutrients; to maintain health;	If groups of nutrients given all must be named	(2)
(ii)	males need more energy than females; 3000 kJ; greater muscle mass; males are likely to be more active/ higher metabolic rate:	Allow reverse for female	(3)
(iii)	iron needed to form haemoglobin; to replace haemoglobin lost during menstruation/red blood cells are lost during menstruation;		(2)
(iv)	vitamin A;		(1)
(v)	In 250g red meat - 2.5 x 2.5 = 6.25; In 200g broccoli - 2 x 0.7 = 1.4; 12 - 7.65; 4.35 (mg);	Full marks for correct final answer ecf for mp3 and mp4	(4)
(b) (i)	bleeding gums/anaemia/bruising/poor wound healing;	Allow other valid symptoms	(1)
(ii)	more fruit/vegetables eaten in diet/increased intake of vitamin C/ vitamin c supplement;		(1)

Total for Question 3 = 14 marks